Predicting Shipment ETA and Delay Propagation Using AI

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1. Objective

For this assignment, I built a first version of a system to predict how many hours late a shipment might arrive. I also looked at which input features the model used most and how delays at different stages were related. I chose synthetic data so I could create the dataset and test the approach within the given time.

2. Dataset and assumptions

I used 300 simulated shipment records. The dataset includes routes, transport modes, scheduled and actual timestamps, port congestion, weather severity, customs workload, and delay values for different stages. Some intermediate timestamps are missing to represent cases where tracking information is incomplete.

This is not real shipping data, so the results depend on the rules used to create it. I would need to check those rules and test the model on historical shipment records before using it for real decisions.

3. Features and model

The model uses origin, destination, transport mode, port congestion, weather severity, customs workload, planned journey duration, and departure month to predict **final_delay_hours**. I did not use actual arrival times or observed stage delays as inputs because that information may not be available when making an initial prediction.

I split the data into 80% for training and 20% for testing. I converted text categories into numeric columns using pandas and filled missing input values using medians from the training data. I trained a Random Forest Regressor and compared it with a simple baseline that predicts the median delay from the training set.

4. Evaluation results

Model / metric	Result	What it means
Median baseline MAE	15.93 hours	Average absolute error from the simple baseline.
Random Forest MAE	8.01 hours	Average absolute error on the test records.
Random Forest RMSE	12.37 hours	Penalizes larger prediction errors more strongly.

The Random Forest had an MAE of 8.01 hours, compared with 15.93 hours for the baseline. That is about a 49.7% reduction in MAE on this test split. This is a useful first result, although it does not tell us how the model would perform on real shipments. The RMSE is higher than the MAE, which suggests that some predictions have larger errors.

5. Feature importance and delay relationships

Severe weather had the highest feature-importance score (about 0.516). Planned journey duration followed at about 0.111, then air transport (0.057), high port congestion (0.052), and departure month (0.034). This shows which features the trained model relied on most; it does not prove that they cause delays.

The average delays in the dataset were 3.32 hours at departure, 11.26 hours at port arrival, and 8.14 hours at customs. Port-arrival delay had a correlation of 0.91 with final delay, while customs delay had a correlation of 1.00. I would check the data-generation code to understand why these relationships are so strong before making claims about real shipping behaviour.

6. Limitations and next steps

- The dataset is simulated and contains only 300 records, so results may change with another train/test split.
- The model predicts final delay from planned information and simulated conditions. It does not independently prove how a delay travels from one stage to another.
- Some predictions are far from the actual delays, especially for shipments with large delays.
- Feature importance shows model reliance, not cause and effect.
- My next step would be to review the simulation rules and, if possible, test the approach on real historical shipment data.

7. Conclusion

This project helped me practise preparing data, selecting features, training a regression model, and evaluating its predictions. The Random Forest performed better than the simple baseline on my synthetic test data. I still need to verify the delay-generation logic and test with real shipment data before treating the results as reliable for practical use.